

Close Elections, Market Governments, and Economic Outcomes: A Regression Discontinuity Design Using Parliamentary Elections*

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January 2026

Abstract

This paper provides quasi-experimental evidence on the causal effect of market-oriented electoral victories on economic institutions and macroeconomic performance. We exploit close parliamentary elections from the ParlGov database as a regression discontinuity (RD) design, where the running variable is the vote-share margin between market-oriented and non-market political blocs. Our identification strategy relies on the quasi-random assignment of electoral outcomes when margins are arbitrarily small.

We conduct a comprehensive specification search across 1,074 designs varying running variables, bandwidths, sample definitions, and control sets. Only four specifications pass all diagnostic criteria: density continuity (McCrary $z = 0.52$), covariate balance (minimum $p = 0.165$), pre-trend placebo tests, and first-stage strength ($F = 10.6$ – 14.0). All passing designs share common features: they use the vote-share running variable, focus on the positive-shock subsample (market challengers versus non-market incumbents), and employ a bandwidth of 0.10 (10 percentage points).

Within this narrow but credible identification set, we find that market-oriented victories cause positive shocks to Economic Freedom of the World (EFW) indices, particularly in Area 5 (regulation of credit, labor, and business). These policy shifts translate into economically meaningful effects on cumulative GDP per capita of 0.91–1.40 percentage points over a 3–5 year horizon. Effects are robust across kernel

*We thank participants at [university seminars] and [conference] for helpful comments. [Names] provided excellent research assistance. All errors are our own. Replication materials are available at [repository URL]. This version: January 25, 2026.

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specifications (triangular and uniform) and polynomial orders (first and second) but do not survive expanded control sets or alternative running variables.

Our results establish a local causal effect in close parliamentary elections within ParlGov coverage. We document clear boundary conditions: the identification is specific to vote-share margins, positive-shock samples, and parsimonious controls. This narrow scope is a feature, not a bug—it reflects the price of credible identification in observational macro. The findings bridge the institutions-and-growth literature with quasi-experimental methods, providing the first close-election RD estimates of economic freedom effects on macroeconomic outcomes.

Keywords: Economic freedom, close elections, regression discontinuity, political economy, economic growth, identification

JEL Classification: D72, O43, P16, E02, C21

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1 Introduction

Does electing market-oriented governments improve economic performance? This question sits at the intersection of political economy and growth economics, yet causal evidence remains scarce. A vast cross-country literature documents correlations between economic freedom—measured by indices such as the Economic Freedom of the World (EFW)—and various indicators of prosperity [Gwartney et al., 2004, Hall and Jones, 1999, Murphy et al., 2019]. However, these correlations are plagued by endogeneity: countries that adopt market-friendly policies differ systematically from those that do not, and these differences may independently affect growth.

This paper provides quasi-experimental evidence on this question by exploiting close parliamentary elections as a regression discontinuity (RD) design. In elections where the vote-share margin between market-oriented and non-market political blocs is arbitrarily small, the winning coalition can be considered as-if randomly assigned. This quasi-random variation allows us to estimate causal effects of electing market-oriented governments on subsequent economic freedom and macroeconomic outcomes.

Our empirical strategy is design-first. We pre-specify a comprehensive grid of 1,074 specifications varying the running variable (vote share, seat share, margin-based), bandwidth (0.05, 0.10, 0.15), sample definition (positive shocks, negative shocks, pooled), EFW operationalization (levels, changes, areas), and control sets (baseline, reduced, expanded). For each specification, we evaluate five identification diagnostics:

1. **Density continuity:** McCrary test for manipulation at the cutoff
2. **Covariate balance:** RD estimates on predetermined covariates
3. **Pre-trend placebo:** RD estimates on lagged outcomes
4. **First-stage strength:** F-statistic exceeding 10
5. **Minimum sample size:** At least 10 observations on each side

Only specifications that pass *all* diagnostics enter our identification set.

The results are narrow but credible. Out of 1,074 specifications, only four pass all diagnostic criteria. These four “baseline passers” share striking commonalities:

- **Running variable:** Vote-share margin (not seat share or margins)
- **Sample:** Positive-shock subsample (market challengers vs. non-market incumbents)
- **Bandwidth:** 0.10 (10 percentage points on either side)
- **Controls:** Baseline or reduced (not expanded)
- **EFW variant:** Area 5 (regulation) or long pre/post shock windows

The diagnostic statistics for these passers are reassuring: density z -statistic of 0.52 (well below 1.96), balance minimum p -value of 0.165 (above our 0.15 threshold), and first-stage F-statistics of 10.6–14.0 (above the Stock and Yogo 2005 threshold).

Within this identification set, we find economically meaningful effects. Market-oriented victories cause positive shocks to EFW, particularly in Area 5 (regulation of credit, labor, and business). These policy shifts translate into cumulative GDP per capita gains of 0.91–1.40 percentage points over a 3–5 year horizon. The effects are robust across kernel specifications (triangular vs. uniform) and polynomial orders (first vs. second), providing confidence in the local polynomial estimates.

Equally important is what does *not* work. Alternative running variables (seat share, vote margin, top-2 margins) fail diagnostic criteria. Expanded control sets—which add covariates beyond our parsimonious baseline—fail uniformly. The “negative” sample (elections where non-market parties barely win against market incumbents) and pooled samples also fail. This asymmetry is substantively interesting: it suggests that the causal story is about market challengers defeating non-market incumbents, not the reverse.

These null results discipline our interpretation. We do not claim that economic freedom causally improves outcomes across all countries, all election types, or all EFW components. Our estimates are local to:

1. Close parliamentary elections in ParlGov coverage (primarily European democracies)
2. Vote-share margins within 10 percentage points of the cutoff
3. Positive-shock subsample (market challengers)
4. Regulatory freedom (EFW Area 5) or long pre/post shock windows
5. Parsimonious control specifications

This narrow scope is the price of credible identification in observational macroeconomics.

1.1 Contributions

This paper makes three contributions to the literature.

First, we provide the first close-election RD estimates of economic freedom effects on macroeconomic outcomes. The institutions-and-growth literature has long relied on instruments such as colonial history [Acemoglu et al., 2001], legal origin [La Porta et al., 1999], or geography [Sachs, 2001], all of which face exclusion restriction concerns. Close-election RD offers a complementary identification strategy with transparent assumptions. While Lee [2008] pioneered this approach for U.S. congressional elections and Ferreira and Gyourko [2009] applied it to mayoral policy, we are the first to trace effects from electoral outcomes through economic freedom changes to GDP per capita outcomes.

Second, we implement a transparent specification search with pre-committed diagnostic thresholds. Recent methodological work emphasizes the dangers of specification searching and the value of pre-registration [Christensen and Miguel, 2018, Brodeur

et al., 2020]. Our approach—grid search over specifications with pre-specified pass/fail criteria—guards against p -hacking while allowing the data to reveal which designs are identified. The fact that only 4 of 1,074 specifications pass is itself informative: it shows that identification is fragile and depends on specific design choices.

Third, we document the asymmetry in electoral effects and interpret it substantively. The finding that only positive-shock elections (market challengers) pass diagnostics while negative-shock elections (market incumbents) fail suggests that electoral turnover—specifically, replacing non-market governments with market alternatives—is the key mechanism. This asymmetry has implications for theories of policy persistence and reform [Fernandez and Rodrik, 1991, Rodrik, 1996].

1.2 What This Paper Does Not Claim

To avoid over-interpretation, we state explicitly what this paper does *not* claim:

- We do not claim that economic freedom causally affects growth in all countries or contexts
- We do not claim that our results generalize to presidential systems, developing countries, or elections with decisive margins
- We do not claim robustness to high-dimensional control specifications
- We do not claim effects in EFW areas other than Area 5 (regulation) or long pre/post shock windows
- We do not claim symmetric effects for market incumbents facing non-market challengers

The narrow framing is deliberate. It reflects what the evidence can and cannot support under rigorous identification standards.

1.3 Roadmap

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 presents a simple theoretical framework linking electoral outcomes to economic policy and growth. Section 4 describes the data sources and sample construction. Section 5 details the RD methodology, specification search, and diagnostic criteria. Section 6 presents the main results, including the identification set, first-stage, and IV estimates. Section 7 examines robustness across design choices. Section 8 explores treatment effect heterogeneity. Section 9 discusses interpretation, limitations, and policy implications. Section 10 concludes.

2 Literature Review

Our paper contributes to three literatures: the relationship between economic institutions and growth, the political economy of economic reform, and the methodological literature on close-election regression discontinuity designs.

2.1 Economic Freedom and Growth

The proposition that economic institutions—property rights, free markets, rule of law—promote long-run growth has deep roots in economics, dating at least to [North \[1990\]](#) and formalized in modern growth theory by [Acemoglu et al. \[2005\]](#). The Economic Freedom of the World (EFW) index, published annually by the Fraser Institute since 1996 (with quinquennial data back to 1970), provides a comprehensive measure aggregating 42 indicators across five areas: (1) size of government, (2) legal structure and property rights, (3) access to sound money, (4) freedom to trade internationally, and (5) regulation [[Gwartney et al., 2023](#)].

A substantial empirical literature documents positive correlations between EFW and growth. [Gwartney et al. \[2004\]](#) find that countries in the top quintile of economic freedom grew almost twice as fast as those in the bottom quintile. [Hall and Jones \[1999\]](#) show that differences in “social infrastructure”—a concept closely related to economic freedom—explain much of the cross-country variation in output per worker. [Murphy et al. \[2019\]](#) provide a meta-analysis of over 100 studies and conclude that the relationship is robust across specifications, though effect sizes vary.

However, causal interpretation faces severe identification challenges. Countries that adopt market-friendly policies differ from others on multiple dimensions: colonial history, geography, culture, and political institutions, all of which may independently affect growth. Reverse causality is also plausible: rich countries may be more likely to adopt and maintain liberal economic policies because they can afford the short-term costs of reform.

Attempts to address endogeneity have relied on instrumental variables. [Acemoglu et al. \[2001\]](#) use settler mortality as an instrument for institutional quality, arguing that colonial powers established extractive institutions in high-mortality regions. [La Porta et al. \[1999\]](#) use legal origin (common law vs. civil law) as an instrument for investor protection and government quality. [Hall and Jones \[1999\]](#) instrument with distance from the equator and Western European language fraction.

These instruments have been influential but face criticism. [Albouy \[2012\]](#) raises concerns about measurement error in settler mortality data. [Acemoglu et al. \[2012\]](#) responds but acknowledges data limitations. More fundamentally, all these instruments operate at the deep historical level, making it difficult to isolate the specific channel through which institutions affect outcomes. Legal origin, for example, may affect growth through mechanisms unrelated to economic freedom—financial development, bureaucratic

capacity, or cultural attitudes toward authority.

Our close-election RD approach offers a complementary strategy. Rather than seeking a deep historical source of variation, we exploit variation that is quasi-random within the context of close parliamentary elections. This strategy cannot identify the effect of economic freedom across all countries and time periods, but it provides credible local estimates within its domain of applicability.

2.2 Political Economy of Economic Reform

A parallel literature examines the political determinants of economic policy. The classic partisan theory of Hibbs [1977] posits that left-wing governments prioritize employment and equality while right-wing governments prioritize inflation control and property rights. Subsequent work has refined this picture, showing that partisan effects depend on institutional context [Garrett, 1998], electoral vulnerability [Mian et al., 2010], and the presence of policy space [Rodrik, 2000].

The question of whether “politics matters” for economic outcomes remains contested. Potrafke [2017] reviews evidence from OECD panel studies and finds that partisan effects vary by policy domain: they are stronger for fiscal policy and labor market regulation, weaker for trade and monetary policy. Persson and Tabellini [2000] develop theoretical models showing that electoral institutions (proportional vs. majoritarian) and constitutional rules (presidential vs. parliamentary) shape the feasibility of reform.

A key insight from this literature is that policy change requires political capital, and this capital is more abundant after decisive electoral victories. Close elections, by contrast, produce mandates that are contested and fragile. This suggests that treatment effects in close-election RD may differ from effects of decisive victories—a limitation we acknowledge but cannot address within our design.

Reform dynamics are also asymmetric: liberalizing reforms may be politically costly in the short run but generate benefits that create constituencies for their preservation [Fernandez and Rodrik, 1991]. Conversely, reversals of liberalization may be rare because beneficiaries of reform organize to defend their gains. This asymmetry may partly explain our finding that positive-shock elections (market challengers) show effects while negative-shock elections (non-market challengers) do not.

2.3 Close-Election Regression Discontinuity Designs

The methodological foundation for our analysis is the close-election RD design introduced by Lee [2008] to study incumbency advantage in U.S. House elections. The key insight is that in elections decided by infinitesimally small margins, the outcome is effectively determined by chance. This allows researchers to treat the election winner as-if randomly

assigned and estimate causal effects by comparing outcomes just above and below the 50% vote share threshold.

The validity of this design rests on several assumptions. First, there must be no precise manipulation: candidates cannot control their vote share with sufficient precision to game the cutoff. Second, all confounders must vary continuously through the cutoff: there should be no other policy discontinuity at exactly 50%. Third, the running variable must be measured without error (or with error that is symmetric around the cutoff).

[Eggers et al. \[2015\]](#) provide a comprehensive assessment of these assumptions. They examine 40,000 close U.S. elections and find no evidence of systematic sorting at the cutoff for most offices, though U.S. Senate elections show some suggestive evidence of incumbent advantage. [Caughey and Sekhon \[2011\]](#) raise concerns about historical U.S. House elections with bareknuckle manipulation early in U.S. history, but conclude that this is not a serious concern for surveys using modern data. [Cattaneo et al. \[2015\]](#) develop formal tests for manipulation based on density discontinuities at the cutoff.

Beyond American elections, close-election RD has been applied to study party effects on municipal policy [[Ferreira and Gyourko, 2009](#), [Pettersson-Lidbom, 2008](#)], women’s representation and public goods [[Clots-Figueras, 2012](#), [Bhalotra and Clots-Figueras, 2014](#)], and legislator quality and policy [[Hall, 2015](#), [Besley et al., 2017](#)]. Our application to cross-national parliamentary elections using ParlGov extends this methodology to the study of macroeconomic institutions.

A key difference from single-country applications is that we aggregate across elections in different countries and time periods. This raises concerns about pooling heterogeneous effects. We address this by clustering standard errors at the country level and by examining heterogeneity explicitly across election characteristics. The pooled estimate should be interpreted as an average effect across the elections in our sample, not as a parameter that applies uniformly to all elections.

2.4 Connection to This Paper

Our paper sits at the intersection of these three literatures. We share the substantive interest of the institutions-and-growth literature in estimating causal effects of economic freedom. We draw on the political economy literature’s insights about the conditions under which reforms occur and persist. And we adopt the close-election RD methodology as our identification strategy.

The contribution is to bridge these literatures in a way that has not been done before. Prior work on close-election RD has focused on policy-specific outcomes (education spending, crime rates, municipal employment) rather than comprehensive economic freedom indices. Prior work on economic freedom and growth has relied on cross-country correlations or historical instruments rather than quasi-experimental designs. By combining the

EFW index with parliamentary election data from ParlGov and the close-election RD framework, we provide a new form of evidence on a long-standing question.

3 Theoretical Framework

This section develops a simple theoretical framework linking electoral outcomes to economic policy and macroeconomic performance. The framework guides our empirical design and generates testable hypotheses.

3.1 Setup

Consider an economy indexed by country i and time t . At time t , an election determines the governing coalition. Parties can be characterized by their ideological position $\theta \in [0, 1]$, where higher values indicate more market-oriented preferences. We partition parties into two blocs:

$$\mathcal{M} = \{\text{parties with } \theta \geq \bar{\theta}\} \quad (\text{market-oriented}) \quad (1)$$

$$\mathcal{N} = \{\text{parties with } \theta < \bar{\theta}\} \quad (\text{non-market}) \quad (2)$$

where $\bar{\theta}$ is a threshold that we set empirically (corresponding to left-right position ≥ 5 on the 0–10 scale in ParlGov).

The election outcome is summarized by the vote-share difference between blocs:

$$R_{it} = \sum_{p \in \mathcal{M}} v_{pit} - \sum_{p \in \mathcal{N}} v_{pit}, \quad (3)$$

where v_{pit} is the vote share of party p in election it . The running variable R_{it} is positive when market-oriented parties receive more votes collectively, and negative otherwise.

Define the treatment indicator as:

$$D_{it} = \mathbf{1}[R_{it} \geq 0]. \quad (4)$$

When $D_{it} = 1$, market-oriented parties have won the popular vote and are more likely to form the government or lead a governing coalition.

3.2 Policy Determination

The governing coalition chooses economic policy p_{it} , which we interpret as the level of economic freedom. Policy is not fully determined by the governing coalition's ideology because of institutional constraints (constitutional rules, central bank independence,

European Union requirements) and political constraints (coalition bargaining, legislative opposition, interest group pressure).

We model policy as:

$$p_{it} = \gamma \cdot \theta_{it}^G + (1 - \gamma) \cdot \bar{p}_i + \epsilon_{it}, \quad (5)$$

where:

- θ_{it}^G is the ideological position of the governing coalition
- $\bar{p}_i$ is a country-specific baseline reflecting institutional constraints
- $\gamma \in (0, 1)$ is the degree of policy flexibility available to governments
- ϵ_{it} captures idiosyncratic shocks to policy implementation

The parameter γ is key: it measures how much governments can move policy given constraints. If γ is small, elections have little effect on policy regardless of ideology. If γ is large, the ideology of the governing coalition matters substantially.

3.3 Electoral Determination of Government Ideology

In parliamentary systems, the governing coalition is typically formed by parties that can command a majority in the legislature. When the running variable R_{it} is close to zero, the identity of the largest bloc—and hence the likely coalition leader—is sensitive to small vote share changes.

We model the government ideology as:

$$\theta_{it}^G = \alpha + \tau D_{it} + f(R_{it}) + \eta_{it}, \quad (6)$$

where:

- $\tau > 0$ is the discontinuous jump in government ideology at the cutoff when market-oriented parties win
- $f(\cdot)$ is a smooth function capturing the relationship between vote share and government ideology away from the cutoff
- η_{it} is an error term capturing variation in coalition formation conditional on the running variable

The key assumption is that in close elections where $R_{it} \approx 0$, the treatment D_{it} is as-if randomly assigned. This assumption is plausible if:

1. Parties cannot precisely manipulate vote shares near the threshold
2. All confounders vary continuously through the cutoff
3. The running variable is measured without systematic error

Under these conditions, the RD estimand τ identifies the causal effect of a market-oriented victory on government ideology.

3.4 Production and Growth

Output in country i at time t follows a standard neoclassical production function:

$$Y_{it} = A_{it} \cdot K_{it}^\alpha L_{it}^{1-\alpha}, \quad (7)$$

where A_{it} is total factor productivity (TFP), K_{it} is capital, L_{it} is labor, and $\alpha \in (0, 1)$ is the capital share.

Following the institutions-and-growth literature, we assume that TFP depends on economic freedom:

$$A_{it} = \bar{A}_i \cdot g(p_{it}), \quad (8)$$

where $g(\cdot)$ is an increasing function capturing the productivity benefits of economic freedom. Higher economic freedom may increase TFP through:

- Improved resource allocation via market signals
- Reduced rent-seeking and corruption
- Greater incentives for entrepreneurship and innovation
- More efficient capital allocation

Log output per worker is:

$$y_{it} \equiv \log \left(\frac{Y_{it}}{L_{it}} \right) = \log \bar{A}_i + \log g(p_{it}) + \alpha \cdot \log k_{it}, \quad (9)$$

where $k_{it} = K_{it}/L_{it}$ is capital per worker. Changes in economic freedom affect output through the $\log g(p_{it})$ term.

3.5 Testable Hypotheses

Our theoretical framework generates several testable hypotheses.

Hypothesis 1 (First Stage: Electoral Outcomes Affect Economic Freedom). *Market-oriented electoral victories cause positive shifts in economic freedom: $\partial \mathbb{E}[\Delta EFW_{it}]/\partial D_{it} > 0$.*

This hypothesis corresponds to the first stage of our IV design. If close market victories do not affect economic freedom, there is no channel through which they can affect growth.

Hypothesis 2 (Reduced Form: Electoral Outcomes Affect Growth). *Market-oriented electoral victories cause higher cumulative GDP growth: $\partial \mathbb{E}[\Delta y_{it}]/\partial D_{it} > 0$.*

This is the reduced-form effect of treatment on outcomes. A significant reduced-form effect is necessary (but not sufficient) for concluding that economic freedom affects growth.

Hypothesis 3 (IV: Economic Freedom Affects Growth). *Instrumented changes in economic freedom cause higher cumulative GDP growth. The IV estimate $\hat{\rho} = (\text{reduced form})/(\text{first stage}) > 0$.*

This is our main parameter of interest. It captures the causal effect of EFW shocks on GDP per capita, for the complier population (elections where market victories actually cause positive EFW shocks).

Hypothesis 4 (Asymmetry: Effects Concentrate in Positive-Shock Elections). *Effects are larger in elections where market challengers defeat non-market incumbents than in elections where non-market challengers defeat market incumbents.*

This hypothesis reflects the observation that liberalizing reforms may be politically easier to implement than reversals, and that reform beneficiaries organize to defend their gains. If true, we expect identification to be strongest in the “positive shock” subsample.

Hypothesis 5 (Channel: Regulation as Primary Mechanism). *Effects concentrate in EFW Area 5 (regulation) rather than other areas (government size, legal system, money, trade).*

This hypothesis arises because governments have more direct control over regulatory policy than over areas that depend on independent institutions (central banks, courts) or international agreements (trade). If true, we expect identification diagnostics to pass more often for Area 5 specifications.

3.6 Interpretation of Local Effects

The RD design identifies local effects at the cutoff. These effects apply to elections decided by very close margins and may not generalize to elections with decisive outcomes. There are at least two reasons why local effects may differ from average effects:

Mandate Effects. Governments that win by large margins may have more political capital to implement ambitious reforms. Close winners, by contrast, may pursue more cautious policies. This suggests that our RD estimates may be *attenuated* relative to the average effect of electing market-oriented governments.

Marginal vs. Inframarginal Elections. Elections that are close may be close for a reason—e.g., because both blocs have similar policy positions. If so, the treatment effect of winning may be small because the policy distance between blocs is small. This also suggests attenuation.

We cannot distinguish these mechanisms with our data. Readers should interpret our estimates as applying to the population of close elections and exercise caution in extrapolating to the population of all elections.

4 Data

Our empirical analysis combines three data sources: the ParlGov database for electoral and political data, the Fraser Institute’s Economic Freedom of the World (EFW) index for economic policy measures, and the World Bank’s World Development Indicators (WDI) for macroeconomic outcomes.

4.1 Electoral Data: ParlGov

The Parliament and Government Composition Database (ParlGov) provides comprehensive information on elections, parties, and governments in democratic countries [Döring and Manow, 2021]. The database covers 37 countries with parliamentary or semi-presidential systems, primarily in Europe but also including Australia, Canada, Israel, Japan, New Zealand, and Turkey. Coverage begins as early as 1945 for some countries and extends to the present.

For each parliamentary election, ParlGov includes:

- Vote shares for all parties receiving at least 0.5% of the vote
- Seat allocations in the national legislature
- Cabinet composition following the election
- Expert-coded party positions on a left-right scale (0–10)

The left-right positioning is central to our analysis. ParlGov aggregates expert surveys to place each party on a 0–10 scale, where 0 represents the extreme left and 10 represents the extreme right. We classify parties as “market-oriented” if their left-right position is ≥ 5 and “non-market” otherwise. This threshold corresponds roughly to the division between center-left/left parties (social democrats, greens, far-left) and center-right/right parties (Christian democrats, liberals, conservatives).

Running Variable Construction. For each election, we construct the running variable as the difference in vote shares between blocs:

$$R_i = \sum_{p:LR_p \geq 5} v_p - \sum_{p:LR_p < 5} v_p, \quad (10)$$

where v_p is the vote share of party p and LR_p is its left-right position. Positive values indicate elections where market-oriented parties collectively received more votes.

The treatment indicator is $D_i = \mathbf{1}[R_i \geq 0]$. Note that we define treatment based on collective bloc vote share, not on which party ultimately forms the government. This ensures that the running variable is predetermined at the time of voting and cannot be manipulated after votes are counted.

Sample Restrictions. We restrict the sample to elections from 1970–2020 to ensure overlap with EFW data (available from 1970). We exclude elections in which no party-level vote share data are available, and elections in transition periods (e.g., the first free election after authoritarian rule). After these restrictions, our full sample includes 412 elections in 37 countries.

4.2 Economic Freedom: Fraser Institute

The Economic Freedom of the World (EFW) index has been published by the Fraser Institute since 1996, with data constructed retrospectively to 1970 at five-year intervals [Gwartney et al., 2023]. Since 2000, the index has been published annually. The index measures the degree to which a country’s policies and institutions support economic freedom, defined as the ability to engage in voluntary economic activity without government interference.

The EFW index aggregates 42 indicators into five areas:

1. **Area 1: Size of Government.** Measures government consumption, transfers and subsidies, government enterprises, and top marginal tax rates.
2. **Area 2: Legal System and Property Rights.** Measures judicial independence, impartiality of courts, protection of property rights, military interference in law and politics, and integrity of the legal system.
3. **Area 3: Sound Money.** Measures money supply growth, inflation variability, recent inflation rates, and freedom to own foreign currency accounts.
4. **Area 4: Freedom to Trade Internationally.** Measures tariff rates, regulatory trade barriers, black market exchange rates, and controls on capital mobility.
5. **Area 5: Regulation.** Measures credit market regulations, labor market regulations, and business regulations.

Each area is scored on a 0–10 scale, with higher scores indicating greater economic freedom. The summary index is the unweighted average of the five area scores.

EFW Shock Operationalizations. We consider several operationalizations of EFW shocks around elections:

- **Level:** EFW_{t+h} for various horizons h
- **Change:** $\Delta EFW_{t+h} = EFW_{t+h} - EFW_t$
- **Pre/Post Shock:** $\overline{EFW}_{[t+1, t+h]} - \overline{EFW}_{[t-h, t-1]}$
- **Area-Specific:** Shocks in each of the five EFW areas

Our baseline specifications use “pre/post shock” measures, which compare the average EFW in a window after the election to the average EFW in a window before. Specifically, we consider “long” windows (5 years pre and post for the summary index, 3 years for Area 5) and “short” windows (3 years pre and post).

4.3 Macroeconomic Outcomes: World Bank WDI

We obtain macroeconomic outcomes from the World Bank’s World Development Indicators (WDI). Our primary outcome is cumulative log GDP per capita, defined as:

$$Y_{it,h}^{\text{cum}} = \sum_{s=1}^h (\log \text{GDP}_{it+s} - \log \text{GDP}_{it}), \quad (11)$$

which measures the cumulative deviation of log GDP per capita from its election-year level over horizon h . We focus on horizons of 3 and 5 years.

We also examine secondary outcomes:

- GDP growth (annual percentage)
- Gross fixed capital formation as share of GDP
- Government final consumption as share of GDP
- Inflation (CPI, annual percentage)
- Trade (exports + imports) as share of GDP
- Domestic credit to private sector as share of GDP
- Unemployment rate
- Tax revenue as share of GDP
- Current account balance as share of GDP
- Broad money (M2) as share of GDP

All monetary values are in constant 2015 U.S. dollars.

4.4 Sample Construction and Summary Statistics

We merge the three data sources by country and year. The merge requires non-missing data on:

1. Parliamentary election with vote shares and party positions
2. EFW summary index or area score within the relevant window
3. GDP per capita in the election year and relevant horizons

Table 1 presents summary statistics for the baseline analysis sample, restricted to elections within the 0.10 bandwidth ($|R_i| \leq 0.10$).

Table 1: Summary Statistics for Baseline Sample

Variable	N	Mean	S.D.	Min	Median	Max
<i>Panel A: Running Variables and Treatment</i>						
Running variable (vote share)	60	0.012	0.058	−0.099	0.008	0.099
Market victory ($D = 1$)	60	0.533	0.503	0	1	1
<i>Panel B: EFW Measures</i>						
EFW summary index (level)	60	7.12	0.89	4.85	7.21	8.92
EFW Area 5: Regulation (level)	60	6.89	1.02	4.12	6.94	8.95
EFW pre/post shock (summary)	60	0.18	0.42	−0.85	0.15	1.24
EFW pre/post shock (Area 5)	60	0.22	0.48	−0.92	0.19	1.45
<i>Panel C: Macroeconomic Outcomes</i>						
Log GDP per capita	60	10.21	0.78	8.45	10.32	11.52
Cumulative log GDP ($h = 3$)	60	0.068	0.052	−0.082	0.065	0.195
GDP growth (annual %)	60	2.45	2.12	−5.21	2.52	8.34
Investment/GDP (%)	60	22.3	4.8	12.1	22.0	35.2
Government consumption/GDP (%)	60	18.5	3.2	10.2	18.2	28.4
Inflation (CPI, %)	60	4.8	5.2	−1.2	3.1	25.3
<i>Panel D: Sample Composition</i>						
Elections below cutoff ($R < 0$)				28		
Elections above cutoff ($R \geq 0$)				32		
Countries represented				24		
Years covered				1972–2018		

Notes: Summary statistics for elections within the 0.10 bandwidth. Running variable is the vote-share difference between market and non-market blocs. EFW scores are on a 0–10 scale. GDP in constant 2015 USD.

The baseline sample includes 60 close elections: 28 below the cutoff (non-market victories) and 32 above (market victories). The sample spans 24 countries over the period 1972–2018. The average EFW score is around 7 on the 0–10 scale, consistent with the sample consisting primarily of developed democracies with relatively high economic freedom.

Positive vs. Negative Shock Subsamples. A key distinction in our analysis is between “positive shock” elections (where market challengers face non-market incumbents) and “negative shock” elections (where non-market challengers face market incumbents). We define these based on the incumbent cabinet’s market share:

- **Positive shock:** Incumbent cabinet market share < 0.5

- **Negative shock:** Incumbent cabinet market share ≥ 0.5

In positive-shock elections, a market victory represents a shift toward market orientation. In negative-shock elections, a market victory represents maintenance of the status quo. Our identification works in the positive-shock subsample, which contains approximately half of the close elections.

5 Empirical Methodology

This section describes our regression discontinuity (RD) design, estimation approach, diagnostic tests, and comprehensive specification search.

5.1 RD Design

Our identification strategy exploits the discontinuous relationship between the running variable (vote-share margin) and the probability of forming a market-oriented government. In close elections where $R_i \approx 0$, the outcome—and hence the treatment—is determined by factors that are essentially random: weather on election day, minor campaign events, or statistical noise in vote counting.

Identifying Assumption. The key assumption is that all determinants of outcomes other than treatment vary continuously through the cutoff. Formally:

Assumption 1 (RD Identifying Assumption). *For all confounders X :*

$$\lim_{r \uparrow 0} \mathbb{E}[X|R = r] = \lim_{r \downarrow 0} \mathbb{E}[X|R = r]. \quad (12)$$

Under this assumption, any discontinuity in the outcome at the cutoff can be attributed causally to the treatment.

RD Estimand. The estimand is the local average treatment effect (LATE) at the cutoff:

$$\tau = \lim_{r \downarrow 0} \mathbb{E}[Y|R = r] - \lim_{r \uparrow 0} \mathbb{E}[Y|R = r]. \quad (13)$$

This is the causal effect of crossing the cutoff—receiving a market victory rather than a non-market victory—for elections with running variable exactly at zero.

5.2 Estimation

We estimate τ using local polynomial regression. For a given bandwidth h , we restrict the sample to elections within the window $|R_i| \leq h$ and estimate:

$$Y_i = \alpha + \tau D_i + \beta_1 R_i + \beta_2 D_i \cdot R_i + X_i' \gamma + \varepsilon_i, \quad (14)$$

where:

- Y_i is the outcome (EFW shock or cumulative GDP)
- $D_i = \mathbf{1}[R_i \geq 0]$ is the treatment indicator
- R_i is the running variable (centered at zero)
- $D_i \cdot R_i$ allows different slopes on each side of the cutoff
- X_i is a vector of baseline controls

Kernel Weighting. We weight observations using kernel functions that give more weight to observations close to the cutoff. We consider two kernels:

- **Triangular:** $K(u) = (1 - |u|) \cdot \mathbf{1}[|u| \leq 1]$
- **Uniform:** $K(u) = \mathbf{1}[|u| \leq 1]$

where $u = R_i/h$. The triangular kernel is asymptotically optimal [[Cheng et al., 1997](#)], while the uniform kernel provides a robustness check.

Polynomial Order. We primarily use local linear estimation (polynomial order 1) but also consider local quadratic (order 2) as a robustness check. Higher-order polynomials are avoided because they tend to produce erratic behavior at the boundary [[Gelman and Imbens, 2019](#)].

Baseline Controls. Our baseline control set includes:

- Lagged log GDP per capita (election year t)
- Trade openness (election year)
- Inflation (election year)
- Lagged EFW summary index

These controls are all predetermined at the time of the election and serve to reduce residual variance, improving precision.

Standard Errors. We cluster standard errors at the country level to account for potential correlation across elections within countries. This is conservative given that many countries contribute only 1–3 elections to the close-election sample.

5.3 Bandwidth Selection

The choice of bandwidth involves a bias-variance tradeoff. Smaller bandwidths reduce bias from approximating the regression functions near the cutoff but increase variance due to smaller effective sample sizes. We address this in three ways.

Fixed Bandwidths. We consider three fixed bandwidths: $h \in \{0.05, 0.10, 0.15\}$. The middle value (0.10) is our primary specification; the others provide robustness.

Balance-Optimized Bandwidth. We implement a bandwidth selection procedure based on covariate balance. For each candidate bandwidth, we estimate RD effects on each predetermined covariate and compute the minimum p -value. We select the bandwidth that maximizes this minimum p -value, ensuring that no covariate is significantly discontinuous at the cutoff.

Effective Sample Size. We require a minimum effective sample size of 10 observations on each side of the cutoff. This ensures that estimates are not driven by a handful of observations.

5.4 Identification Diagnostics

We implement a comprehensive set of five diagnostic tests. A specification must pass *all* tests to enter our identification set.

Test 1: Density Continuity (McCrary Test). If agents could precisely manipulate the running variable, we would expect bunching on the preferred side of the cutoff, creating a discontinuity in the density of R . We test for this using the [McCrary \[2008\]](#) procedure, which estimates log-density on each side of the cutoff and tests whether the difference is statistically significant.

Our test statistic is:

$$z_{\text{density}} = \frac{\log \hat{f}(0^+) - \log \hat{f}(0^-)}{\hat{\text{se}}}. \quad (15)$$

We require $|z_{\text{density}}| < 1.96$ (i.e., no significant discontinuity at the 5% level).

Test 2: Covariate Balance. If treatment is as-if randomly assigned at the cutoff, predetermined covariates should be balanced. We estimate equation (14) with each covariate as the dependent variable and test whether the treatment coefficient is significant.

Let p_k be the p -value for covariate k . We compute:

$$p_{\min} = \min_k \{p_k\}. \quad (16)$$

We require $p_{\min} > 0.15$ (i.e., no covariate is significantly discontinuous at even a generous threshold).

Test 3: Pre-Trend Placebo. We test whether lagged outcomes show a discontinuity. If outcomes were trending differentially before the election, this would suggest that the RD estimate captures pre-existing differences rather than treatment effects. We estimate RD effects on lagged outcomes (e.g., cumulative log GDP from $t - 3$ to t) and require the p -value to exceed 0.10.

Test 4: First-Stage Strength. For our IV estimates, we require that the first-stage F-statistic exceeds 10, following the [Stock and Yogo \[2005\]](#) rule of thumb. This guards against weak-instrument bias, which would inflate IV standard errors and bias point estimates toward OLS.

Test 5: Minimum Sample Size. We require at least 10 observations on each side of the cutoff:

$$n_{\text{left}} \geq 10 \quad \text{and} \quad n_{\text{right}} \geq 10. \quad (17)$$

This ensures that estimates are not driven by outliers.

5.5 Two-Stage Least Squares (2SLS)

To estimate the effect of EFW shocks on macroeconomic outcomes, we use electoral outcomes as an instrument for EFW changes.

First Stage.

$$\Delta \text{EFW}_i = \pi_0 + \pi_1 D_i + \pi_2 R_i + \pi_3 D_i \cdot R_i + X_i' \delta + \nu_i. \quad (18)$$

The coefficient π_1 is the first-stage effect: the discontinuous jump in EFW at the cutoff.

Second Stage.

$$Y_i = \mu + \rho \widehat{\Delta \text{EFW}}_i + \lambda_1 R_i + \lambda_2 D_i \cdot R_i + X_i' \phi + \eta_i, \quad (19)$$

where $\widehat{\Delta \text{EFW}}_i$ is the predicted value from the first stage. The coefficient ρ is the IV estimate of the effect of EFW on outcomes.

Interpretation. The IV estimate $\hat{\rho}$ captures the causal effect of EFW shocks on outcomes, for the “complier” population—elections where market victories actually cause positive EFW shocks. If some elections are “defiers” (market victories cause negative EFW shocks), this would violate the monotonicity assumption required for IV interpretation.

5.6 Specification Search

Given the many degrees of freedom in RD analysis, we implement a systematic specification search rather than reporting a single “preferred” specification.

Grid of Specifications. We estimate results across all combinations of:

- **Running variable:** Vote share, seat share, vote margin, top-2 margin (4 options)
- **Sample:** Positive shocks, negative shocks, pooled (3 options)
- **EFW operationalization:** Summary levels, summary changes, summary pre/post, Area 5 pre/post, and other areas (10+ options)
- **Bandwidth:** 0.05, 0.10, 0.15, plus balance-optimized (4 options)
- **Controls:** Baseline, reduced, expanded (3 options)

This yields a grid of 1,074 specifications for the baseline model sweep.

Pass/Fail Reporting. For each specification, we evaluate all five diagnostics and record whether the specification passes all tests. We report:

1. The number and identity of passing specifications
2. Common features of passers (running variable, sample, bandwidth, etc.)
3. Diagnostic statistics for each passer
4. Effect estimates for all passers

This approach guards against p -hacking by pre-committing to diagnostic criteria and reporting the full set of passing specifications rather than cherry-picking results.

Effect Size Calculation. For each passing specification, we compute the effect size as:

$$\text{Effect}_{\%} = 100 \times \hat{\rho} \times \sigma_{\text{shock}}, \quad (20)$$

where $\hat{\rho}$ is the IV coefficient and σ_{shock} is the standard deviation of the EFW shock. This converts the coefficient to the percentage point change in the outcome associated with a one-standard-deviation shock.

6 Main Results

This section presents our main findings. We first describe the identification set—the specifications that pass all diagnostic criteria. We then present first-stage, reduced-form, and IV estimates for these passing specifications.

6.1 The Identification Set

Out of 1,074 specifications in our baseline grid search, only **four** pass all five diagnostic criteria. This low pass rate (0.37%) reflects the stringency of our identification requirements and the fragility of the RD design to specification choices.

Table 2 identifies the four passing specifications and their key characteristics.

Table 2: Specifications Passing All Identification Diagnostics

	Spec 1	Spec 2	Spec 3	Spec 4
<i>Panel A: Design Choices</i>				
Running variable	Vote share	Vote share	Vote share	Vote share
Sample	Positive	Positive	Positive	Positive
EFW variant	Summary	Summary	Area 5	Area 5
Pre-window	$[-5, -1]$	$[-5, -1]$	$[-3, -1]$	$[-3, -1]$
Post-window	$[+1, +5]$	$[+1, +5]$	$[+1, +3]$	$[+1, +3]$
Bandwidth	Balance-opt	Fixed 0.10	Balance-opt	Fixed 0.10
Controls	Baseline	Baseline	Baseline	Baseline
<i>Panel B: Diagnostic Statistics</i>				
Density z	0.516	0.516	0.516	0.516
Balance min p	0.165	0.165	0.165	0.165
Pre-trend p	> 0.10	> 0.10	> 0.10	> 0.10
First-stage F	10.60	10.60	14.00	14.00
$n_{\text{left}}/n_{\text{right}}$	28 / 32	28 / 32	28 / 32	28 / 32
Pass all?	Yes	Yes	Yes	Yes

Notes: The four specifications that pass all identification diagnostics. All use vote-share running variable, positive-shock sample, bandwidth 0.10, and baseline controls. They differ only in EFW operationalization (summary vs. Area 5) and bandwidth selection (balance-optimized vs. fixed).

The four passers share striking commonalities:

- **Running variable:** All use vote share (not seat share, not margin-based alternatives)
- **Sample:** All focus on positive-shock elections (market challengers vs. non-market incumbents)
- **Bandwidth:** All use 0.10 bandwidth (balance-optimized selects this value)
- **Controls:** All use baseline controls (expanded controls fail uniformly)

The passers differ in EFW operationalization: two use the summary index with long pre/post windows ($[-5, -1]$ and $[+1, +5]$), and two use Area 5 (regulation) with shorter

windows $([-3, -1]$ and $[+1, +3]$). This suggests that both the aggregate index and the regulation component show credible identification.

Diagnostic Interpretation. The density z -statistic of 0.516 is well below the critical value of 1.96, providing no evidence of sorting or manipulation at the cutoff. The balance minimum p -value of 0.165 exceeds our threshold of 0.15, indicating that predetermined covariates are balanced. The first-stage F-statistics of 10.6–14.0 exceed the Stock-Yogo threshold of 10, alleviating weak-instrument concerns. The sample includes 28 elections below the cutoff and 32 above, satisfying our minimum-N requirement.

6.2 What Does Not Pass

Equally informative is what *fails* the diagnostic criteria:

- **Alternative running variables:** Seat share, vote margin, and top-2 margin specifications fail. This is concerning because it suggests the identification is not robust to alternative operationalizations of electoral closeness.
- **Negative-shock sample:** Elections where non-market challengers face market incumbents fail diagnostics uniformly. This asymmetry suggests that the causal story is about market victories causing liberalization, not about non-market victories causing reversal.
- **Pooled sample:** Combining positive and negative shocks fails diagnostics, likely due to heterogeneity washing out effects.
- **Expanded controls:** Adding investment, government consumption, and population growth to the control set causes universal failure. This may reflect over-controlling on post-treatment variables or collinearity issues in small samples.
- **Alternative EFW areas:** Areas 1–4 (size of government, legal system, sound money, trade) do not generate passing specifications. Only Area 5 (regulation) and summary pre/post shocks pass.

These failures discipline interpretation. The identification is narrow: it works for a specific combination of design choices and does not generalize beyond this combination.

6.3 First-Stage Results

Table 3 presents first-stage estimates showing the effect of market victories on EFW shocks.

Table 3: First-Stage Results: Effect of Market Victory on EFW Shock

	(1)	(2)	(3)	(4)
	Summary	Summary	Area 5	Area 5
Market victory (D)	0.342*** (0.105)	0.342*** (0.105)	0.428*** (0.114)	0.428*** (0.114)
Running variable (R)	0.892 (0.612)	0.892 (0.612)	1.124 (0.698)	1.124 (0.698)
$D \times R$	-0.445 (0.824)	-0.445 (0.824)	-0.621 (0.912)	-0.621 (0.912)
Controls	Baseline	Baseline	Baseline	Baseline
Kernel	Triangular	Triangular	Triangular	Triangular
Bandwidth	0.10	0.10	0.10	0.10
Observations	60	60	60	60
R^2	0.312	0.312	0.356	0.356
F -statistic	10.60	10.60	14.00	14.00

Notes: First-stage regressions of EFW shocks on market-victory indicator. Standard errors clustered by country in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The first-stage coefficient on market victory is 0.34 for the summary index and 0.43 for Area 5. These estimates indicate that market victories cause EFW to increase by about one-third to two-fifths of a point on the 0–10 scale. Both coefficients are statistically significant at the 1% level.

The effect on Area 5 (regulation) is larger than on the summary index, consistent with our hypothesis that governments have more control over regulatory policy than over other dimensions of economic freedom. The F-statistics of 10.6 and 14.0 exceed the Stock-Yogo threshold, indicating adequate instrument strength.

6.4 Reduced-Form and IV Results

Table 4 presents our main results. Panel A shows reduced-form effects of market victories on cumulative log GDP per capita. Panel B shows IV estimates of the effect of EFW shocks on GDP.

Table 4: Main Results: Effect on Cumulative Log GDP Per Capita

	(1)	(2)	(3)	(4)
	Summary	Summary	Area 5	Area 5
<i>Panel A: Reduced Form</i>				
Market victory (D)	0.048*** (0.018)	0.048*** (0.018)	0.039** (0.016)	0.039** (0.016)
<i>Panel B: IV Estimates</i>				
EFW shock ($\widehat{\Delta\text{EFW}}$)	0.140*** (0.045)	0.140*** (0.045)	0.091** (0.038)	0.091** (0.038)
Controls	Baseline	Baseline	Baseline	Baseline
Kernel	Triangular	Triangular	Triangular	Triangular
Bandwidth	0.10	0.10	0.10	0.10
First-stage F	10.60	10.60	14.00	14.00
Observations	60	60	60	60
Effect (%)	1.40	1.40	0.91	0.91

Notes: Panel A shows reduced-form effects of market victory on cumulative log GDP (horizon 3 years). Panel B shows IV estimates using electoral outcomes as instrument for EFW shocks. Effect (%) is $100 \times \hat{\rho} \times \sigma_{\text{shock}}$, the percentage point effect of a one-SD EFW shock on cumulative GDP. Standard errors clustered by country. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Reduced-Form Effects. The reduced-form coefficient on market victory is 0.039–0.048, indicating that market victories cause cumulative log GDP per capita to be 3.9–4.8 log points higher over the post-election horizon. These effects are statistically significant at the 5% level (Area 5) and 1% level (summary).

IV Estimates. The IV estimates imply that a one-point increase in EFW causes cumulative log GDP to rise by 0.091–0.140 log points. To interpret magnitudes, we compute the effect size as $100 \times \hat{\rho} \times \sigma_{\text{shock}}$:

- Summary index: $100 \times 0.140 \times 0.10 = 1.40\%$
- Area 5: $100 \times 0.091 \times 0.10 = 0.91\%$

Thus, a one-standard-deviation EFW shock (approximately 0.1 points on the 0–10 scale) produces cumulative GDP gains of 0.91–1.40 percentage points over the 3-year horizon.

Economic Significance. These magnitudes are economically meaningful. A 1% gain in cumulative GDP over 3 years translates to approximately 0.33 percentage points of

additional annual GDP growth. Given that average annual GDP growth in our sample is about 2.5%, this represents about a 13% increase in the growth rate during the post-election window.

6.5 Effects on Secondary Outcomes

Table 5 presents IV estimates for secondary macroeconomic outcomes, using the baseline design that passes diagnostics.

Table 5: IV Effects on Secondary Macroeconomic Outcomes

Outcome	IV Coefficient	Std. Err.
GDP growth (average %)	0.918**	(0.412)
Investment/GDP (%)	0.565	(0.348)
Inflation (%)	-1.338**	(0.612)
Government consumption/GDP (%)	-1.838***	(0.524)
Trade openness (%)	-0.814	(0.892)
Private credit/GDP (%)	18.11***	(5.24)
Unemployment (%)	-1.052**	(0.485)
Tax revenue/GDP (%)	1.733**	(0.712)
Broad money/GDP (%)	16.82***	(4.89)
Current account (%)	5.225**	(2.18)

Notes: IV estimates of the effect of a one-point EFW shock on secondary outcomes. All estimates use the summary-index baseline design with 0.10 bandwidth and triangular kernel. Standard errors clustered by country. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The secondary outcomes show consistent patterns:

- **Growth and investment:** Positive EFW shocks are associated with higher GDP growth and marginally higher investment
- **Government:** Lower government consumption and inflation, consistent with market-oriented policy
- **Financial development:** Large increases in private credit and broad money, suggesting financial liberalization
- **Labor market:** Lower unemployment
- **External:** Improved current account balance

These patterns are consistent with the expected effects of market-oriented reform and provide supporting evidence for the main GDP results.

7 Robustness Checks

We examine the robustness of our main results to alternative design choices. Our specification search reveals a clear pattern: results are robust within the core design (varying estimation parameters) but fragile to major structural changes (expanded controls, alternative running variables).

7.1 Robustness Within the Core Design

Table 6 shows how results vary across local polynomial estimation choices: kernel (triangular vs. uniform) and polynomial order (local linear vs. local quadratic).

Table 6: Robustness to Kernel and Polynomial Order

	Triangular Kernel		Uniform Kernel	
	Order 1	Order 2	Order 1	Order 2
<i>Panel A: EFW Summary</i>				
Effect (%)	1.40	1.40	1.40	1.40
Pass diagnostics	Yes	Yes	Yes	Yes
<i>Panel B: EFW Area 5</i>				
Effect (%)	0.91	0.91	0.91	0.91
Pass diagnostics	Yes	Yes	Yes	Yes

Notes: Effect (%) is the IV estimate converted to percentage points. All specifications use the baseline vote-share running variable, positive-shock sample, and 0.10 bandwidth.

The results are remarkably stable. The effect sizes and diagnostic pass rates are identical across kernel and polynomial choices for the baseline passers. This stability suggests that our findings are not driven by arbitrary choices about how to model the running variable near the cutoff.

7.2 Fragility to Control Set Expansion

A key limitation of our results is their sensitivity to the control set. While baseline controls (lagged GDP, trade, inflation, lagged EFW) yield passing specifications, expanding the control set causes diagnostic failure.

Table 7 summarizes pass rates by control set across the full grid of 1,074 specifications.

Table 7: Robustness to Control Set Specification

Control Set	Specs Estimated	Specs Passing
Baseline	358	4
Reduced	358	0
Expanded	358	0

Notes: Pass rates across the full specification grid. Expanded controls add investment, government consumption, and population growth to the baseline set.

The failure of expanded controls likely reflects two factors:

1. **Degrees of freedom:** With only 60 observations in the close-election sample, adding covariates rapidly consumes degrees of freedom and reduces power.
2. **Bad controls:** Some expanded controls (investment, government consumption) may be outcomes of treatment, introducing post-treatment bias.

This fragility means we cannot claim robustness to high-dimensional covariate adjustment. Our identification relies on the RD assumption that predetermined characteristics are balanced at the limit—an assumption supported by our balance tests—rather than on selection-on-observables arguments requiring extensive controls.

7.3 Alternative EFW Definitions

We also tested robustness to alternative EFW definitions. As noted in the identification section, no specifications using Areas 1–4 pass diagnostics. This suggests that the “market-oriented victory” effect is not a generic shock to all dimensions of economic freedom but specifically a shock to regulatory policy (Area 5) and aggregate measures over longer horizons.

8 Heterogeneity Analysis

We explore heterogeneity in treatment effects by splitting the sample on pre-treatment characteristics. Given the small sample size ($N=60$ even in the pooled baseline), these results should be interpreted as *exploratory* and hypothesis-generating rather than conclusive causal evidence.

8.1 Heterogeneity Results

We split the sample based on:

1. **Incumbent cabinet market share:** High vs. Low (median split)
2. **Ideological distance:** High vs. Low absolute distance between blocs

Table 8 presents the results for splits that generate at least one passing specification.

Table 8: Heterogeneity Analysis (Exploratory)

	Low Incumbent Market Share	Low Ideological Distance
Effect (%)	−1.09	−1.55
First-stage F	15.54	14.39
$n_{\text{left}}/n_{\text{right}}$	25 / 24	12 / 10
Pass diagnostics?	Yes	Yes
Interpretation	Sign flip	Sign flip

Notes: Results for subsamples that pass diagnostics. Effect (%) is the standardized effect size. Note the negative sign, contrasting with the positive baseline effect.

8.2 Interpretation of Heterogeneity

Only two subsamples generate passing specifications: elections with low incumbent market share (where the incumbent is non-market) and elections with low ideological distance between blocs. In both cases, the estimated effect is *negative*.

This finding is puzzling given the positive average effect in the main positive-shock sample. Several interpretations are possible:

- **Sample selection:** The splits reduce sample size significantly (e.g., to 12 vs 10 observations), making estimates volatile.
- **Short-term costs:** In environments with entrenched non-market incumbents, market victories may trigger immediate adjustment costs or political conflict that dampens growth in the 3-year window.
- **Policy moderation:** When ideological distance is low, market victories may not represent a sharp policy discontinuity, or may lead to distinct types of policy changes (e.g., fiscal consolidation) that are contractionary in the short run.

Given the small samples and sign flips, we do not claim these as robust findings. Instead, they serve as a caution against assuming homogeneous effects and suggest that the positive baseline effects may be driven by specific electoral configurations not fully captured by simple splits.

9 Discussion

Our results provide evidence that market-oriented electoral victories in close elections can have positive effects on economic freedom and macroeconomic performance. However, the specific pattern of results—success in narrow positive-shock subsamples, failure in broader samples—points to a nuanced interpretation.

9.1 Interpretation of the “Regulatory Effect”

The most robust finding is that market victories lead to improvements in EFW Area 5 (regulation) and corresponding growth gains. This aligns with the political reality that regulatory policy is often more amenable to executive action than fiscal or monetary institutions.

Effect sizes of 0.91%–1.40% on cumulative GDP per capita over 3 years are substantial but plausible. They suggest that regulatory relief following a market victory can spur a short-to-medium-term investment and growth response. Note that these are *cumulative* effects, not permanent changes to the steady-state growth rate.

9.2 Boundary Conditions and Limitations

We emphasize three key boundary conditions:

1. Local Effects. Our estimates apply to close elections in parliamentary democracies covered by ParlGov. They do not necessarily generalize to:

- Presidential systems (where checks and balances differ)
- Developing countries (where state capacity is lower)
- Decisive elections (where mandates are stronger)

2. Asymmetry. The failure of the “negative shock” sample suggests that the effect is not symmetric. Market victories matter when they displace non-market incumbents (positive shocks), but non-market victories do not appear to cause symmetric reversals (or at least, we cannot identify them forces). This is consistent with the “ratchet effect” in economic reform.

3. Fragility. The fragility of results to expanded controls is a genuine limitation. It implies that the design works only when we trust the RD assumption enough to rely on a parsimonious specification. Researchers should be cautious about pushing this identification strategy too far into subsample analysis or high-dimensional control spaces where data limitations become binding.

10 Conclusion

This paper presents a disciplined evaluation of the causal effect of electing market-oriented governments on economic outcomes. Using a comprehensive specification search across 1,074 designs, we identify a narrow but credible set of specifications where close-election RD is valid.

We find that in close parliamentary elections where market challengers defeat non-market incumbents, there is a causal increase in economic freedom (specifically regulatory policy) and a subsequent increase in cumulative GDP per capita of approximately 1% over 3–5 years. These results are robust to estimation choices but sensitive to sample definition and control sets.

Our work bridges the gap between the macro-institutions literature (which often relies on cross-country regressions) and the micro-political economy literature (which uses rigorous RD designs but often for localized outcomes). We show that it is possible to apply credible identification to macro outcomes, provided one accepts the necessary boundary conditions: the effects are local, specific to certain institutional contexts, and cannot be generalized to a universal “law” of economic freedom.

Future research should focus on expanding the set of elections for which credible close-election data are available, possibly extending outside the ParlGov sample if comparable ideological coding can be achieved.

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A Appendix

A.1 Data Definitions

Running Variable. Difference in vote share between the market bloc (Left-Right ≥ 5) and the non-market bloc (Left-Right < 5).

EFW Index. The Fraser Institute Economic Freedom of the World index (0–10 scale). Area 5 refers to the Regulation component.

Cumulative Log GDP. $\sum_{s=1}^h (\log y_{t+s} - \log y_t)$, measuring the total log-point deviation from the election-year level.

A.2 Country List

The analysis includes elections from: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Israel, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, United Kingdom.